
MAM3000W 3AL

University of Cape Town

Final Examination June 2022

Department of Mathematics and Applied Mathematics

Time: 120 minutes **Available marks:** 95 **Full marks:** 90

Instructions

1. Answer all questions in the lined answer booklets. **Start each question on a new page. Number each page and put your student number or EMPL ID on the top of each page.**
2. Show all work and reasoning unless otherwise instructed.
3. This exam consists of 5 questions. Make sure your question paper is complete.
4. If you've read the instructions write the name of your favourite mathematician on the inside cover of your booklet. You will receive one extra mark for doing so.

Good luck! ✨

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1. (a) State the First Isomorphism Theorem.

   (b) Let $G = \mathrm{GL}_2(\mathbb{R})$ be the group of invertible 2×2 matrices over $\mathbb{R}$. We define a subgroup of G as follows:

   $$S = \{A \in G : \det(A) = \pm 1\}.$$

   Prove that $G/S \cong \mathbb{R}^{>0}$, where $\mathbb{R}^{>0}$ is the multiplicative group of positive real numbers.

   [5 + 10 = 15 marks]

2. (a) State the Third Isomorphism Theorem.

   (b) Prove the Third Isomorphism Theorem.

   [5 + 15 = 20 marks]

3. Let G be a group. Recall that $\mathrm{Aut}(G)$ is the group of automorphisms of G (i.e. the set of isomorphisms $\varphi : G \to G$ equipped with the operation of composition). Consider the group action $* : \mathrm{Aut}(G) \times G \to G$ defined by $\varphi * x = \varphi(x)$.

   (a) Prove that $*$ is a group action.

   (b) Let $x \in G$. Prove that every element in the orbit of x has the same order as x.

   [5 + 10 = 15 marks]

4. Let G a group of order $p^n m$ where $n \geq 1$ and p is a prime such that $p \nmid m$. Suppose that H is a Sylow p-subgroup of G.

   (a) Explain why H is a Sylow p-subgroup of $N_G(H)$ and why it is the *only* Sylow p-subgroup of $N_G(H)$.

   (b) Let $a \in G$ such that $|a| = p^m$ where $1 \leq m \leq n$. Prove that if $aHa^{-1} = H$, then $a \in H$.

   Hint: Every p-subgroup is contained in a Sylow p-subgroup.

   [10 + 10 = 20 marks]

5. (a) State the Third Sylow Theorem.

   (b) State the definition of a simple group. Give an example of a group that is simple and an example of a group that is not simple.

   (c) Show that any group of order $30 = 2 \times 3 \times 5$ is not simple.

   [5 + 5 + 15 = 25 marks]